

# Joshua Satya Cetron, Ph.D. | *Resume*

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*Data scientist, applied statistician, research methodologist, cognitive neuroscientist*

## Education

**Ph.D., Psychology, Harvard University, Cambridge, MA, May 2023**

*M.A. in Psychology also conferred in March 2021*

- Advisors: Mina Cikara, Ph.D. (primary advisor), Joshua Greene, Ph.D.
- Presidential Scholar, Harvard Graduate School of Arts and Sciences
- National Science Foundation Graduate Research Fellow

**B.A., Neuroscience, *Summa Cum Laude*, Dartmouth College, Hanover, NH, June 2016**

- GPA: 3.92, Phi Beta Kappa, Rufus Choate Scholar (top 5% of class, 2 years)
- High Honors Award, Neuroscience Honors Thesis (advisor: David Kraemer, Ph.D.)
- James O. Freeman Presidential Scholar and Kaminsky Family Fund Award Grantee

## Professional Experience

**Data Science Specialist, Harvard University Institute for Quantitative Social Science**

*Cambridge, MA, June 2023 - Present*

- Consultant and expert collaborator advising on the development and implementation of methods and statistical analyses for cutting-edge scientific research at Harvard and MIT
- Special expertise: experimental methods, advanced regression modeling, high-dimensional representational analysis, programming and data visualization in R

**Data Management Expert, CAFE RCC for Health & Extreme Weather**

*Harvard University and Boston University, June 2023 - Present*

- Senior member of the Data Management Function for the NIH-funded CAFE Research Coordinating Center at Boston University School of Public Health and Harvard T.H. Chan School of Public Health
- Built and maintain a multi-platform system for sharing data and code, analysis tutorials, and translational research expertise among the health and extreme weather research community of practice

**Owner and Expert Consultant, Satya Data Solutions, LLC**

*Somerville, MA, July 2025 - Present*

- Expert consultant in data science, research analysis, and statistical modeling
- Developing, scoping, and directing statistical analysis, data management, and multi-modal data harmonization for researchers and industry professionals
- Translational research expertise across disciplines including the social and behavioral sciences, healthcare and biological sciences, cognitive science and artificial intelligence

**Doctoral Researcher, Harvard University**

*Cambridge, MA, September 2018 - May 2023*

- Built and implemented a research program on the impact of intergroup bias on evidence-based learning and the mental representations of factual and opinion statements
- Developed new methods advancing neuroscience, psychology, philosophy, and statistics

**Teaching Fellow in Doctoral Statistics, Harvard University**

*Cambridge, MA, September 2020 - December 2024*

- Taught advanced statistics courses for Harvard doctoral students in the social sciences alongside Senior Lecturer in Statistics Patrick Mair, Ph.D.
- Sample topics: Advanced regression modeling (GLMMs, GAMs, robust and resistant regression, mixture models), Bayesian inference, cluster analysis, hidden Markov models, regularization (LASSO, Ridge)
- Three-time awardee of the Derek Bok Center Certificate of Distinction in Teaching

## Skills and Expertise

- **Statistical consulting expertise:** 3 years of professional consulting on over 150 unique cases supporting statistical analysis, methods development, data management, research planning, experimental design
- **Advanced expertise in R** for statistical modeling and data visualization
- **Advanced regression modeling** (frequentist and Bayesian): Linear and generalized linear fixed-effects, mixed-effects, and additive modeling, mixture regression, robust and quantile regression
- **Multivariate cluster analyses:** Hierarchical clustering, density-based clustering, K-means
- **Multivariate analyses to process and characterize high-dimensional data:** representational similarity analysis (RSA), multivariate pattern analysis of neuroimaging data (MVPA), multidimensional scaling (MDS) and unfolding analysis, principal components analysis (PCA), support vector machine (SVM) classifiers, informational network analysis
- **Other programming expertise:** Python, shell scripting, HPC batch computing (incl. slurm and PBS), web-scraping, RMarkdown, Quarto, and jupyter notebook
- **Additional skills:** Fluent in Spanish; lifelong musician (25+ years of multi-instrumental training and performance)

## Additional Work Experience

### **Honors Thesis and Research Advisor, Harvard University**

*Cambridge, MA, 2019 - 2022*

- Advised and mentored multiple award-winning undergraduate student researchers

### **Researcher and Lab Manager, Cognitive Neuroscience of Learning Lab, Dartmouth College**

*Hanover, NH, 2013 - 2018*

- Designed and conducted neuroimaging, behavioral, and classroom-based research to explore the multivariate neural representations of STEM learning
- Developed novel analytical methods to characterize knowledge representations in the brain
- Authored 6 academic publications, 5 of which were published within a 2-year period

### **Director, Dartmouth Outing Club (DOC) First-Year Trips Program**

*Hanover, NH, 2015 - 2016*

- Directed the largest College outdoor orientation program in the country: 1000 incoming students participating in 139 five-day, student-led outdoor trips across New Hampshire's White Mountains
- Interviewed and hired a 20-person Directorate including 1 Assistant Director and 19 team leaders
- Selected and managed a student volunteer staff of 350+ (278 trip leaders + 56 support crew members on 6 teams) from an applicant pool of 600+

### **Laboratory Intern and Research Assistant, Emory University Transplant Centers**

*Atlanta, GA, 2010 - 2012*

- Assisted with data collection, specimen management, and statistical analysis in the laboratory of Leslie Kean, M.D., Ph.D. studying graft-versus-host disease in bone marrow transplant patients

### **Program Facilitator, Pearson Seminar on Youth Leadership (PSYL)**

*Lester B. Pearson United World College of the Pacific, Victoria, B.C., Canada, 2010 & 2011*

- Designed and implemented a month-long summer leadership program on social justice, global citizenship, environmental sustainability, and community-building for 100 high school students from 20+ countries, alongside 16 other facilitators and 8 program coordinators